

Meridian: A Technical Autopsy

A forensic RAG measurement framework — engineering post-mortem

Validated on LegalBench-RAG (ContractNLI, CUAD, MAUD, PrivacyQA; 194 queries per corpus; combined ~11,524-chunk SAC index) and a single non-legal transfer test (NFCorpus, BEIR).

1. Thesis: why deterministic measurement is the deliverable

The pipeline is not the product. The measurement layer is.

A retrieval pipeline is a moving target — chunking, fusion, routing, and synthesis all change the numbers, and they change them in ways that interact. The hard problem in RAG evaluation is not generating a number; it is *attributing* a number. When correctness rises 8 points, you must be able to say whether the pipeline got better or the ruler moved. If those two causes are confounded, every result is suspect and no experiment compounds, because you can never be sure the previous "win" was real.

Meridian's answer is strict layer separation. **Layer 1** is a deterministic span-overlap taxonomy that scores *retrieval* against ground-truth character offsets with no LLM in the loop. It is the trust anchor: the same input always produces the same classification, so when a Layer-1 number moves, the pipeline moved — full stop. **Layer 2** is an LLM-judged pair of metrics (correctness and faithfulness) that scores *answer* quality, with a pinned judge model, run as a separate, clearly-labeled step. The two layers never share code paths and never feed each other; a Layer-2 verdict can never enter the Layer-1 taxonomy, and Layer-1 classifications never steer Layer-2.

The reason this matters is not architectural elegance. It is that a wrong number produced by a silent bug is *indistinguishable from a right one* — both are plausible floats. The only defense is an instrument rigorous enough to contradict itself: a deterministic ruler calibrated against an external published baseline, against which an implausible result stands out. The proof that the framework works is not the headline gains. It is that **four silent measurement bugs were caught before they shipped wrong numbers** — each one caught by a verify-before-trust check that the layer separation made possible. Section 4 is the centerpiece of this report for exactly that reason.

2. Headline numbers

The headline is the **config-stack delta** measured on the combined index (all four corpora pooled into one ~11,524-chunk SAC collection, matching the paper's benchmark regime). Both arms share the same index, so this is a controlled, method-level comparison (Finding 45).

	Arm 0 (SAC + RRF)	Arm 1 (best config)	Δ
Correctness (avg, span-informed judge)	60.5%	68.3%	+7.9pp
Faithfulness (avg, holistic groundedness)	91.7%	95.9%	+4.2pp

Arm 1 = SAC chunking + CC fusion (per-corpus α) + always-on document routing (top-3) + unconditional LLM selector + no reranker + single-shot synthesis, on DeepSeek-v4-flash. Per-corpus breakdowns and retrieval metrics are in Section 5; the external system-vs-system comparison is in Section 6.

3. The measurement layer

3.1 Layer 1 — the deterministic span taxonomy

Layer 1 classifies each query's retrieval against ground-truth character spans into one of six mutually-exclusive failure types, evaluated along a fixed priority chain with **DRM checked first**:

- **DRM — Document Retrieval Miss.** The correct *document* never reached the final context window. Checked first because it dominates: a wrong document guarantees a wrong answer (Section 8), so it must short-circuit the chain before any finer-grained label applies.
- **CBF — Chunk Boundary Failure.** The right document was retrieved but the chunk boundary cut across the answer span, so the retrieved chunk has zero character overlap with the ground-truth span.
- **ICR — Incomplete Chunk Retrieval.** The right region was retrieved but only partially covers the ground-truth span.
- **OVR — Over-Retrieval.** The retrieved chunk contains the ground-truth span but is substantially larger than it (a region-quality, not a miss, signal).
- **SGP — Span Group Partial.** On multi-span answers, some but not all of the required spans were captured. This is scored **per span, never on a merged character set** — merging spans before scoring would mask partial multi-span coverage, which is the exact failure SGP exists to detect.
- **OK — clean retrieval.** All required spans captured within well-sized chunks.

Per-span (not merged) scoring is a hard rule, not a preference: the DRM-first priority chain means that when DRM is high, the later branches are starved, so the taxonomy's *shape* shifts as the pipeline changes. This is by design — but it is also why Layer 1 must never steer control flow. A classification that moves with config cannot also be a controller, or a measurement bug becomes indistinguishable from a behavior bug. The classifier additionally emits a HIGH/MEDIUM/LOW confidence, a nearest-alternative category, and a boundary-distance evidence dict, all additive — the labels themselves stay deterministic.

3.2 Layer 2 — correctness and faithfulness

Correctness uses a span-informed judge (DeepSeek-flash, temp 0, thinking disabled, pinned and held constant across the campaign): does the answer capture the golden-span information, judged semantically rather than by string match? This replaced an earlier affirmative-only judge ("did the system say YES"); the two are **not comparable** — span-informed runs ~15–17pp higher by design, and every reported delta holds the judge constant (Finding 24).

Faithfulness is holistic groundedness (the RAGAS / RAG-triad definition): is each answer claim supported by the *full* retrieved context (all top-8 chunks the model saw), not just the chunk the model chose to cite (Findings 26, 30). Judging against the full context is deliberate — judging a claim only against its own cited snippet is partly blind to the OK-but-incorrect failure faithfulness exists to instrument. A separate, demoted **citation-precision** diagnostic (claim vs its own cited chunk) is retained only where citations are meaningful (extractive/legal), and is never the faithfulness headline.

The two Layer-2 metrics are **orthogonal** (Finding 26): a DRM query scores faithfulness 1.0 (the model faithfully reports what the *wrong* document says) while scoring correctness 0. Correctness catches wrong-document and wrong-conclusion failures; faithfulness catches hallucination and contradiction. They measure two different failure modes and are never collapsed into one.

3.3 Non-contamination

The "no LLM judges" rule applies to Layer 1 only — the deterministic span-overlap taxonomy, the trust anchor. Answer correctness legitimately requires a judge and lives entirely in Layer 2 (Finding 20). The two are reported separately and never contaminate each other. This was clarified after the rule was initially over-applied; the resolution is precise scoping, not loosening.

3.4 Ruler calibration (Finding 44)

The foundational check: replicate the published LegalBench-RAG baseline stack *through Meridian's own measurement layer* and compare to the paper's numbers (arXiv 2408.10343, Table 5; RCTS 500-char, text-embedding-3-large, dense-only). Formula alignment was confirmed first (any-character-overlap = hit; $P@k = |\cap|/|\text{retrieved_k_chars}|$; $R@k = |\cap|/|\text{gt_chars}|$; k-unit = chunks; 194-query split), then the empirical reproduction:

Corpus	P@1 Δ vs paper	R@8 Δ vs paper	Status
MAUD	+0.39pp	+2.29pp	Calibrated
CUAD	+1.65pp	+2.85pp	Calibrated
PrivacyQA	+2.91pp	-2.14pp	Near-calibrated (at boundary)
ContractNLI	+5.88pp	+12.9pp	Diverges (benchmark-file provenance)

PrivacyQA uses the identical 194-query set with no sampling, so its ~2–3pp divergence establishes the **embedding-drift floor** — the irreducible noise from text-embedding-3-large not being byte-frozen since the paper's Aug-2024 version. Three of four corpora land inside it. ContractNLI's larger residual was localized to benchmark-file provenance (its `contractnli.json` carries `query_id` fields absent from the paper's schema and a different 194-query sample; divergence persists even on the 166 shared-doc queries). That residual is almost certainly provenance, not a ruler bug — but it is not *directly* proven (proving it would require re-running the paper's full generation pipeline), so ContractNLI's external comparison is caveated throughout.

4. The four caught bugs

Each of these produced a *plausible* output. None crashed in the place that mattered. Each was caught only because a verify-before-trust check had something to contradict.

Bug 1 — The Pro model-ID alias (every "Pro" test was flash)

(a) **Mechanism.** Phase-8 synthesis comparisons used `PRO_MODEL = "deepseek-chat"`. That string is a legacy provider alias that routes to `deepseek-v4-flash` (non-thinking) — the *same* model Arms 0 and 1 already ran. Every "Pro" arm in the project's history was actually flash against flash.

(b) Why it was invisible. The harness saved only the *requested* model string into the record's `model` field, never the API response's served model. The record said `deepseek-chat`, which is exactly what was requested — internally consistent, nothing to flag. And flash produces fluent, well-cited legal answers, so a flash-vs-flash comparison looked like a real Pro-vs-flash one: the resulting -1.7pp aggregate delta sat squarely inside the synthesis noise band and read as a clean finding ("Pro is null on the hard cases"). A wrong number that lands in the expected range is the hardest kind to catch.

(c) How verification caught it. Three independent signals: (1) DeepSeek's own docs state `deepseek-chat` currently routes to `deepseek-v4-flash` non-thinking (alias retiring 2026-07-24); (2) the served model was never logged — the gap itself was the tell; (3) the **\$0.00 line on the deepseek-v4-pro billing tier** after a run that supposedly used Pro. A real Pro run cannot cost zero on the Pro tier. The billing line was the smoking gun.

(d) Fix. Corrected the ID to `deepseek-v4-pro` and added served-model logging — `synthesize()` now extracts `served_model` from the instructor raw response and writes it as a distinct field (Finding 42). A mandatory model-tier verification protocol followed: `served_model` must match the intended tier, latency signature must be consistent (Pro ran 2.3–4.2x flash), and the billing line must be non-zero. The corrected Pro run (verified 776/776 `served_model = deepseek-v4-pro`) showed Pro indistinguishable from flash within the variance band (-1.7pp correctness avg, +0.9pp faithfulness — both inside noise), at 2.3–4.2x latency. Flash stays the default; the conclusion held, but only after the comparison was real.

Bug 2 — BM25 / dense channel mismatch on the combined index

(a) Mechanism. On the combined index, the dense channel searched the pooled mini-document SAC chunks while the BM25 channel searched the *full* corpus parquets (96K chunks for CUAD). The two fusion channels were operating over different ID spaces.

(b) Why it was invisible. A channel mismatch does not necessarily error — fusion still merges two ranked lists and emits a plausible final ranking. On ContractNLI it silently *corrupted* results without raising anything; the numbers came out wrong but well-formed. The mismatch only became visible because, on CUAD specifically, the size disparity *crashed* the run.

(c) How verification caught it. The CUAD crash forced a postmortem, and the postmortem traced the crash to the channel size mismatch — which then revealed that ContractNLI had been silently corrupted by the same root cause. The visible failure exposed the invisible one.

(d) Fix. BM25 was filtered to the same mini-document set as the dense channel, channel-matching both sides of fusion. CUAD and ContractNLI were both re-run.

Provenance note: DECISIONS.md records this as "dense searched mini-doc chunks, BM25 searched full parquets (96K for CUAD)." The specific "4,496 mini-doc chunks" figure from the brief is not in the log; described above without that number.

Bug 3 — The zero-span offset bug ((0,0) defaults → near-zero P@k/R@8)

(a) Mechanism. Qdrant payloads do not store span offsets — spans live in the parquets. The combined-index builder defaulted the missing span fields to `(0, 0)`. Every P@k/R@8 was then computed against an empty `(0, 0)` ground-truth span, driving all retrieval metrics to near-zero.

(b) Why it was invisible. This is the most dangerous of the four: near-zero retrieval scores are a *plausible result, not a crash*. CUAD reported R@8 ≈ 0 while routing recall was 100%. To a reader without an external anchor, "this config

retrieves badly" is a perfectly ordinary thing for a number to say — there is no exception, no stack trace, nothing structurally wrong with the output. It is simply a believable wrong answer.

(c) How verification caught it. Sanity-check against the Finding 44 calibration. The calibrated CUAD R@8 is ~0.81; a near-zero reading coexisting with 100% routing recall is a flat contradiction — routing demonstrably *finds* the right document on every query, so recall against it cannot be ~0. The calibration anchor turned a believable number into an impossible one.

(d) Fix. Spans are now looked up from the parquets and the metrics were recomputed from the saved records and spot-checked. Real CUAD R@8 came back at **0.814** — exactly in the calibrated range.

Provenance note: the log states "All P@k/R@8 computed as ~0"; the specific "0.049" figure in the brief is consistent with that "~0" but is not separately recorded in DECISIONS.md.

Bug 4 — Chunk-size inconsistency (MAUD 2048-char vs others 512-char)

(a) Mechanism. The four corpora's baseline SAC collections use two different chunk sizes: ContractNLI, PrivacyQA, and CUAD use 512-char / 128-overlap, while MAUD uses 2048-char / ~512-overlap. The blanket project claim that "our SAC uses ~2048-char chunks" was wrong for three of the four corpora (Finding 46).

(b) Why it was invisible. This is not a runtime bug — every chunk size produces valid, internally-consistent numbers. The inconsistency is inert *internally* and only becomes a defect in the **external** comparison, where the paper's baseline uses RCTS 500-char. MAUD's 2048 vs 500 is a 4x granularity confound that mechanically deflates P@k (larger chunks lower per-chunk precision). Nothing in MAUD's own numbers looks off; the confound is only exposed when you line MAUD up against an external 500-char baseline.

(c) How verification caught it. Baseline-verification / calibration work surfaced the size while auditing the external comparison stack. MAUD's larger chunks were also retroactively understood (Finding 28) to have *accidentally helped* by holding whole multi-span merger clauses intact in one retrievable unit — a pragmatic choice (collection size: 512-char on MAUD's ~350K-char merger docs would yield ~192K chunks vs ~45K) that turned out beneficial, and that would have been silently confounded if compared externally.

(d) Fix. The inconsistency is documented, and **MAUD is excluded from the external P@k/R@k multiplier comparison** (Section 6). The three 512-char corpora remain valid against the paper's 500-char baseline.

5. Results

Config-stack delta — Arm 0 (RRF) → Arm 1 (best config), same combined index

Corpus	Correctness Arm 0 → Arm 1 (Δ)	Faithfulness Arm 0 → Arm 1 (Δ)
ContractNLI	62.9% → 71.1% (+8.2pp)	89.1% → 94.1% (+5.0pp)
PrivacyQA	49.0% → 55.7% (+6.7pp)	94.4% → 97.9% (+3.5pp)
CUAD	61.9% → 73.7% (+11.8pp)	92.1% → 96.0% (+3.9pp)
MAUD	68.0% → 72.7% (+4.7pp)	91.3% → 95.5% (+4.2pp)
Average	60.5% → 68.3% (+7.9pp)	91.7% → 95.9% (+4.2pp)

This is the controlled, method-level claim: SAC is held constant on both sides, so the delta isolates CC fusion + routing + selector over the RRF baseline. Every per-corpus correctness delta (+4.7 to +11.8pp) exceeds the ± 2 –4pp synthesis variance band (Section 8); these are real. Arm 1 cost 2,077–4,813 tokens/query at 5.6–8.4s latency.

Retrieval metrics — Arm 1 (real spans, recomputed after Bug 3 fix)

Corpus	Arm 0 P@1	Arm 1 P@1	Arm 0 R@8	Arm 1 R@8
ContractNLI	0.152	0.422	0.702	0.810
PrivacyQA	0.068	0.297	0.462	0.579
CUAD	0.027	0.394	0.605	0.814
MAUD	0.008	0.270	0.670	0.783

6. External comparison (system vs system)

Meridian Arm 1 against the published LegalBench-RAG baseline (arXiv 2408.10343, Table 5; RCTS 500-char, text-embedding-3-large, dense-only, no reranker), on the three un-confounded corpora:

Corpus	Meridian P@1	RCTS P@1	Mult	Meridian R@8	RCTS R@8	Mult
ContractNLI [caveat]	0.422	0.066	6.4x	0.810	0.250	3.2x
PrivacyQA	0.297	0.144	2.1x	0.579	0.424	1.4x
CUAD	0.394	0.020	19.7x	0.814	0.317	2.6x

Framing (non-negotiable). These multipliers compare **Meridian's full stack** (SAC + CC + hybrid + routing + selector, on voyage-4) against the paper's **bare baseline** (RCTS + dense-only, on text-embedding-3-large). The advantage bundles embedder generation + hybrid-vs-dense + fusion + routing — it is *not* a method-alone claim. The honest statement is "our optimized system vs their published baseline," never "our method is Nx better." **MAUD is excluded** (2048-char vs

500-char, 4x confound — Bug 4). **ContractNLI is caveated** (benchmark-file provenance differs from the paper's, and the paper's own RCTS ContractNLI baseline has a wide, unstable range — Finding 44). The clean comparisons are CUAD and PrivacyQA.

7. NFCorpus transfer (retrieval only)

The first non-legal test (NFCorpus, BEIR medical IR; 3,633 docs, 323 queries, nDCG@10). Two things were learned, and the boundaries on both are important (Finding 47).

The genuine finding is the routing characterization. Routing *monotonically hurt* on NFCorpus across all fusion weights (OFF 0.3969 > k=10 0.3331 > k=5 0.2726 > k=3 0.2272 at $\alpha=0.1$). Because NFCorpus has dispersed relevance — many documents per query are relevant — the top-k hard filter discards relevant documents. This confirms the project's standing claim that **routing is a concentrated-relevance technique whose benefit tracks document-discrimination difficulty** (consistent with Findings 23 and 45): it helps where one document answers the query (legal) and hurts where relevance is spread across many (medical). The system self-identified this via the sweep and self-disabled routing.

The retrieval result is real but bounded. Meridian's best config (CC $\alpha=0.1$, routing OFF) reached nDCG@10 = 0.3988 vs its own RRF baseline 0.3430 — CC fusion contributing +5.6pp, comparable to its +7.9pp on legal, confirming CC is a general (not domain-specific) improvement. That 0.3988 **beats the classic BEIR baselines**, including the BM25+cross-encoder reranker (0.350), BM25 (0.325), contriever (0.328), and ColBERT (0.305). Caveats: this is system-vs-system (our hybrid + a 2024 embedder vs single-method + 2021-era models); these are *classic* baselines, not modern SOTA dense retrievers; and it is a **base-retrieval transfer only**. BEIR provides document-level relevance, not character spans, so **Layer 1 — the actual contribution — could not run there**. The measurement framework's transferability to a non-legal corpus with span annotations is *not yet demonstrated*.

8. Limitations, and what Meridian is not

Affirmative-only correctness. The legal corpora's ground-truth answers are affirmative (the queried provision exists). Correctness therefore measures recall on provisions that *exist*; it does not test the false-positive rate (correctly answering NO when a provision is absent), because the benchmark has no negative cases. Real-deployment correctness would differ.

Variance bands gate every quality claim. Grounded-correctness has a ± 2 – 4 pp synthesis variance floor from LLM nondeterminism alone (measured via an accidental flash-vs-flash duplicate, Finding 40); faithfulness is tighter at ± 0.6 pp; R@8 carries a ± 0.50 pp floor from voyage embedding nondeterminism (Finding 4). Any delta below these bands is noise and is never narrated as a result. The config-stack deltas (Section 5) clear the band; the Pro-vs-flash delta did not.

The comprehension floor is real and unreachd. ~ 2 – 3 genuine grounded-but-wrong comprehension cases per corpus (the model has correct, grounded evidence and reasons to the wrong answer) resisted every tested lever — section chunking, hierarchical chunking, cross-reference graph, multi-query, the repaired agentic loop, CoT, decomposition, and a (verified) stronger model (Findings 27–39). The framework's value here is precisely that it *characterized* this floor rather than papering over it. Most of the residual that looked like comprehension turned out to be within-document **access miss** (GT span retrievable but ranked below top-8); the LLM selector recovers a slice of it (Finding 39).

ContractNLI routing degrades on the combined index. With cross-corpus distractors present, ContractNLI routing recall drops to 76%, and 47/194 queries (24%) get zero chunks from the correct document. This is a genuine system limitation on topically-homogeneous corpora, not a measurement artifact (Finding 45).

Phase-9 verification is not a faithfulness metric. Its NLI step false-positives on legal negation ("shall not", "no obligation") at 0.20–0.33 on MAUD; it stays in the pipeline as a conservative safety check, but the holistic Layer-2 judge is the faithfulness headline (Finding 41).

What Meridian is *not*. It is not an autonomous research agent and not a general-purpose production RAG bot. Several of its winning choices are explicitly benchmark-scoped: CC fusion suits calibrated single-index corpora and is *not* a universal production default (RRF remains the robust production choice for noisy, heterogeneous-system retrieval); routing is concentrated-relevance only; the strongest model tier is unjustified at this scale. The contribution is the measurement layer — a deterministic ruler disciplined enough to catch its own four silent bugs — not any single pipeline component.